

Title Placeholder: Physics-Informed Diffusion for OD-Conditioned Trajectory Generation

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Abstract

Accurate simulation of human mobility is fundamental to resilient urban planning and the development of Digital Twins. However, current modeling approaches face a critical validity gap: deterministic baselines fail to capture the stochastic nature of individual choices, while emerging generative models often suffer from “macroscopic shrinkage,” producing diverse but physically unrealistic trajectories that underestimate urban velocity and displacement. Here, we propose a **Physics-Informed Residual Diffusion** framework that effectively reconciles microscopic stochasticity with macroscopic physical consistency. By decomposing trajectory generation into a deterministic low-frequency anchor and a stochastic high-frequency residual, our approach decouples the learning of physical scale from the modeling of uncertainty. We show on a city-scale Shenzhen taxi dataset that this hierarchical formulation resolves the shrinkage artifact, achieving state-of-the-art coverage (best-of- K) while restoring correct motion statistics. This work offers a scalable, data-driven engine for valid urban simulation, bridging the divide between deep generative modeling and transportation engineering.

1 Introduction

Understanding and simulating human mobility is a cornerstone of modern urban engineering, underpinning applications from traffic signal timing to mobility “Digital Twins” that support scenario analysis and decision-making under uncertainty (Wang et al., 2022). For these systems to be effective, they require trajectory models that are **valid** simulations of reality—capturing not just the average flow, but the realistic stochasticity of driver choices.

However, existing modeling approaches face a critical validity gap. Classic deterministic models (e.g., L2 regression) effectively treat every driver as an optimal agent, collapsing all uncertainty into a single mean path. This behavior filters out the “tail events”—such as rare route choices or

sudden deviations—that are precisely what risk-aware planners need to model. Conversely, emerging generative models (like pure diffusion) promise diversity but can suffer from a severe physical artifact we explicitly diagnose in this work: **macroscopic shrinkage**. While they generate diverse routes, the simulated vehicles systematically fail to maintain realistic speeds and displacements, weakening their utility for engineering-grade simulation.

A recurring theme in the transportation Digital Twin literature is that the term “twin” is sometimes used loosely. In transport planning and infrastructure management, survey evidence emphasizes that “Digital Twins” are expected to support continual data integration and a service/connection layer that keeps the digital representation synchronized with the physical system (Nag et al., 2025; Yan et al., 2023). From this perspective, trajectory generation is not only a prediction task but also a data-model component that can enrich simulation inputs for what-if analysis.

To bridge this gap between **stochastic coverage** and **physical realism**, we propose a pragmatic, data-driven framework: Physics-Informed Residual Diffusion. Instead of relying on expensive road-graph engineering or complex kinematic constraints, we demonstrate that a simple residual decomposition—using a frozen deterministic baseline as a low-frequency anchor—can effectively calibrate the generative model. This provides a clear, low-cost path to upgrading traffic simulators from “mean-value estimators” to “physically consistent scenario generators”.

Our approach and contributions. We develop a reproducible pipeline for **dt-fixed** trajectory generation ($\Delta t = 30s$) and compare: (i) a deterministic L2 regression baseline, (ii) a conditional diffusion trajectory generator (Ho et al., 2020), and (iii) a physics-conditioned diffusion model that injects a *navigation field* (a train-only, leakage-free local mean-flow prior) as additional conditioning. We implement a strict data contract to prevent leakage (train-only normalization and `nav_field`) and adopt an evaluation protocol for generative models including best-of- K , distribution metrics, and macroscopic motion statistics. We further provide a systematic diagnosis of the shrinkage issue and report ablations on training-time macro constraints.

The remainder of the report introduces the data and methods, presents quantitative and spatial results, and discusses limitations and future directions for city-scale mobility modeling.

2 Related Work

2.1 Data-Driven Mobility Simulation for Digital Twins

Urban mobility “Digital Twins” are increasingly framed as connected cyber-physical loops rather than static simulation models. In transport planning and infrastructure management, recent surveys stress clear boundary conditions that distinguish a full twin from a digital model or digital shadow,

typically requiring continual data integration, a service layer, and an explicit connection between physical and digital components (Nag et al., 2025; Yan et al., 2023). Within this broader ecosystem, mobility twins have been discussed as an architectural stack with data, models, and services, together with practical challenges around scalability, security, and deployment (Wang et al., 2022; Ferré-Bigorra et al., 2022). These trends align with the smart-city Digital Twin literature, where large-scale bibliometric evidence highlights rapid growth and the central role of AI, IoT, and privacy/security considerations (El-Agamy et al., 2024).

The development of mobility Digital Twins therefore requires high-fidelity simulation engines capable of reproducing the complex, stochastic behaviors of human mobility agents. Traditionally, mobility simulation has relied on Agent-Based Models (ABM) and microscopic kinematic rules (Helbing and Molnár, 1995). While interpretable, these systems often require labor-intensive manual calibration of parameters and struggle to scale to city-wide heterogeneity. Consequently, data-driven approaches have emerged as a scalable alternative, learning agent behaviors directly from massive trajectory logs.

Early data-driven methods typically formulated trajectory prediction as a deterministic regression problem, utilizing recurrent architectures (RNNs, LSTMs) to model temporal dependencies (Alahi et al., 2016), or Transformer-based mechanisms to capture long-range interactions (Zhou et al., 2022; Shi et al., 2022). However, deterministic regression inherently minimizes point-wise mean error, leading to predictions that collapse towards a single conditional average. In the context of Digital Twins, this “mode collapse” is critical: it filters out stochastic tail events—such as rare route choices or delays—that are essential for robust risk assessment and scenario planning.

To recover this lost stochasticity, generative sequence models have been increasingly adopted. Variational Autoencoders (VAEs) and Conditional GANs have been applied to sample diverse future trajectories (Gupta et al., 2018). More recently, Denoising Diffusion Probabilistic Models (DDPMs) have demonstrated superior capability in capturing complex, multimodal distributions without the training instability of GANs (Ho et al., 2020; Zhu et al., 2023). Innovations such as Leapfrog Diffusion (Mao et al., 2023) and guidance-based control (Wang et al., 2024) further enhance the controllability of these generators. Yet, a persistent challenge in applying these generative models to city-scale simulation is *physical validity*: generated trajectories can exhibit “macroscopic shrinkage” or kinematic inconsistencies, undermining the fidelity required for engineering-grade Digital Twins.

From a system perspective, several recent efforts demonstrate how mobility simulation, real-time traffic streams, and AI modules can be combined into an operational Digital Twin workflow. Open and scalable urban mobility operating systems have been proposed to support city-scale experimentation and visualization (Yeon et al., 2023), and real-time synergy between traffic data streams and simulation has been emphasized for virtualizing motorway dynamics (Kušić et al., 2023). Related work also highlights Digital Twin designs at corridor/freeway scales (Fu et al., 2022)

and motivates DT services through resilience and adaptability objectives in transportation systems (Feng et al., 2023). In this context, data-driven trajectory generation can be viewed as a modular AI capability that enriches the data-model layer of a mobility twin, complementing broader DT+AI integration patterns discussed in the infrastructure literature (Wu et al., 2022).

2.2 Physics-Informed Inductive Biases in Urban Computing

To ensure that data-driven models respect the physical laws of urban dynamics, Physics-Informed Machine Learning (PIML) has become a key research frontier. In the context of mobility, “physics” refers to constraints imposed by road network topology, traffic flow dynamics, and kinematic limits. Explicit structural constraints have been integrated into diffusion models via graph-based priors (Shi et al., 2024) or road-network-constrained generation (Wei et al., 2024). Beyond explicit constraints, inductive biases derived from vector fields or partial differential equations (PDEs) are also employed to guide the learning process (Bastek et al., 2025).

Our work aligns with this philosophy but adopts a specific focus on *macroscopic flow consistency*. Rather than imposing rigid kinematic equations which may limit inference speed, we employ a non-parametric navigation field—derived from historical flow statistics—as a soft physical prior. This approach anchors the generative process in the empirical reality of the city’s flow dynamics, providing a balanced solution for constructing the “behavioral layer” of an urban Digital Twin.

3 Data and Methods

To address the tension between multi-modal coverage and macroscopic realism, we develop a rigorous experimental pipeline centered on explicit temporal semantics and physical priors. This section details the mathematical problem setup, the dt-fixed preprocessing protocol, and the hierarchical model architectures proposed to recover physical consistency at city scales.

3.1 Problem Setup and Data (dt-fixed)

We study **OD-conditioned trajectory generation with a known destination** (KnownDestination). Trajectories are represented in a grid coordinate system. Let $\mathbf{p}_t \in \mathbb{R}^2$ denote the 2D position at step t (grid indices in $[y, x]$ order), and define step displacement (velocity) as

$$\mathbf{v}_t = \mathbf{p}_{t+1} - \mathbf{p}_t. \quad (1)$$

We use a fixed sampling interval $\Delta t = 30\text{s}$ (dt-fixed), so $\mathbf{v}_t$ has explicit time semantics and is proportional to physical velocity. We use an observation window length $H = 8$ and prediction

horizon $F = 12$ (4 minutes of history and 6 minutes of future). Denote the observed history as $\mathbf{X} = \{(\mathbf{p}_t, \mathbf{v}_t)\}_{t=-H+1}^0$ and the future velocities as $\mathbf{Y} = \{\mathbf{v}_t\}_{t=1}^F$. Given a global context vector $\mathbf{c}$ (time features and OD endpoints), our goal is to model the conditional distribution

$$p_{\theta}(\mathbf{Y} \mid \mathbf{X}, \mathbf{c}). \quad (2)$$

At inference, we sample K future trajectories (by sampling K future velocity sequences) to represent multi-modality.

Dataset. We use the Shenzhen taxicab GPS dataset (Zhang et al., 2015). We preprocess trajectories into grid space and enforce a strict, leakage-free data contract.

Grid rasterization. We discretize the city bounding box (latitude $[22.45, 22.85]$, longitude $[113.75, 114.65]$) into a $H_g \times W_g = 400 \times 800$ grid. Positions are stored as grid coordinates $\mathbf{p}_t = [y_t, x_t]$.

dt-fixed resampling. We resample each trajectory to a fixed interval $\Delta t = 30\text{s}$ and drop segments with large gaps or insufficient length. After resampling, the split sizes are train=97,222, val=21,103, test=21,309 trajectories.

Leakage-free strict products. All normalization statistics and the navigation field prior are computed using the **train split only**. This prevents information leakage from validation/test into priors or scaling constants.

Normalization and conditioning. We normalize positions and velocities using train-only statistics. The global conditioning vector includes (i) time features (hour-of-day, day-of-week) and (ii) origin and destination (OD) endpoints, both normalized in the same coordinate system.

3.2 Model Framework

We compare a deterministic regression baseline, a conditional diffusion generator, and a physics-conditioned diffusion variant. To address the macroscopic shrinkage observed in Phase B v1.0, we additionally adopt a residual formulation (v1.1) that combines a frozen deterministic prior with stochastic diffusion residuals. Figure 1 summarizes this decomposition.

Deterministic baseline (L2 regression). The baseline predicts a single future velocity sequence by minimizing L2 error. As a deterministic model, it corresponds to $K = 1$ at inference.

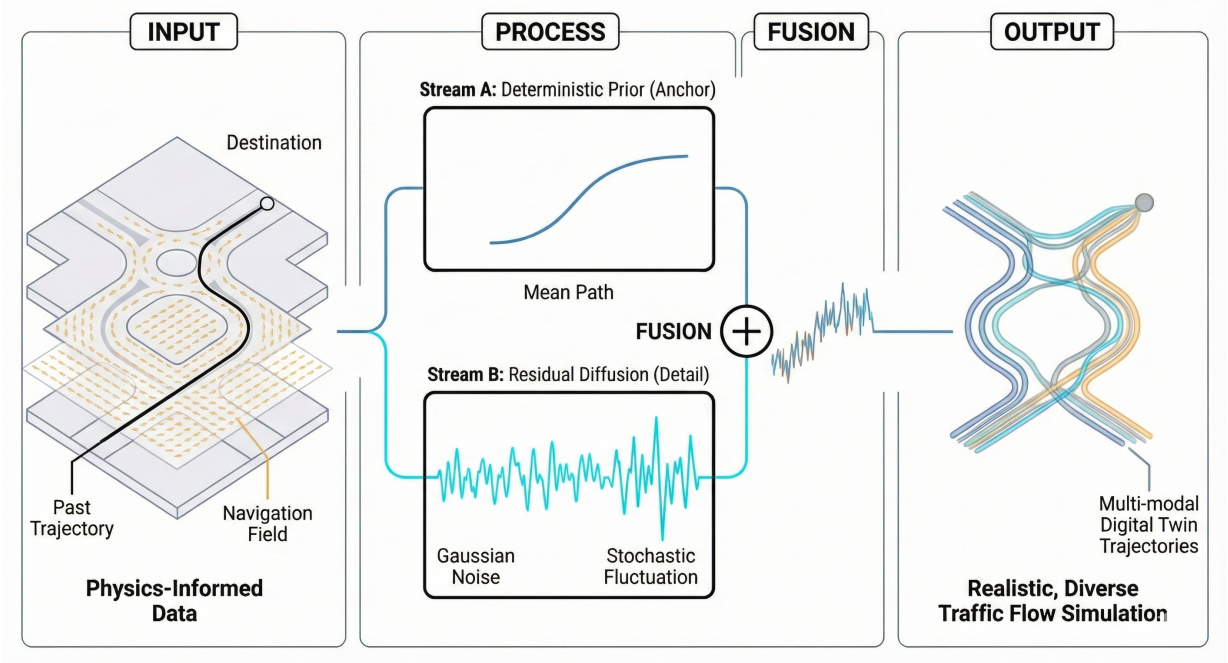


Figure 1: Framework overview. We combine a deterministic low-frequency prior (anchor) with a stochastic residual diffusion model (detail) under OD conditioning and a train-only navigation-field prior. The fusion yields multimodal trajectories that retain macroscopic realism while capturing diverse route-level variation.

Conditional diffusion trajectory generator. We adopt a denoising diffusion probabilistic model (DDPM) (Ho et al., 2020) to sample future velocity sequences conditioned on history and OD context.

Physics-conditioned diffusion (navigation field). We extend the diffusion model by injecting an additional, non-parametric prior in the form of a navigation-field patch around the current position (defined below).

Residual diffusion (v1.1, frozen deterministic prior). In Phase B we observe a systematic “shrinkage” failure mode in pure diffusion: sampled trajectories tend to under-estimate macroscopic displacement (low MSD/Rog). To decouple low-frequency scale from stochastic variation, we adopt a residual formulation:

$$\mathbf{v}_{1:F} = \mathbf{v}_{1:F}^{\text{prior}} + \mathbf{v}_{1:F}^{\text{res}}, \quad \mathbf{v}_{1:F}^{\text{prior}} = f_{\text{base}}(\mathbf{X}, \mathbf{c}), \quad (3)$$

where f_{base} is a frozen deterministic baseline (trained under the same dt-fixed protocol), and the diffusion model learns the conditional distribution of residual velocities $p_{\theta}(\mathbf{v}^{\text{res}} \mid \mathbf{X}, \mathbf{c})$. At inference, we sample residuals and add them back to the prior to obtain full future velocities.

Algorithm 1 Reverse diffusion sampling (future velocities)

[1]

history $\mathbf{X}$, context $\mathbf{c}$, horizon F , diffusion steps T sample $\mathbf{y}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ **for** $t = T, T-1, \dots, 1$ **do** $\epsilon \leftarrow \epsilon_\theta(\mathbf{y}_t, t, \mathbf{c})$ $\mathbf{y}_{t-1} \leftarrow \text{DDPM_STEP}(\mathbf{y}_t, \epsilon, t)$ **end for****return** $\mathbf{y}_0$ (sampled future velocities)

Conditional diffusion (DDPM). We model the clean future velocity sequence as a matrix $\mathbf{y}_0 \in \mathbb{R}^{F \times 2}$, and define a forward noising process that gradually perturbs $\mathbf{y}_0$ into $\mathbf{y}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Using a variance schedule $\{\beta_t\}_{t=1}^T$ with $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$, the forward process admits the closed form

$$q(\mathbf{y}_t | \mathbf{y}_0) = \mathcal{N}\left(\mathbf{y}_t; \sqrt{\bar{\alpha}_t} \mathbf{y}_0, (1 - \bar{\alpha}_t) \mathbf{I}\right). \quad (4)$$

The reverse process is learned with a noise predictor $\epsilon_\theta(\mathbf{y}_t, t, \mathbf{c})$ and optimized using the simplified objective

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{\mathbf{y}_0, t, \epsilon} \left[\|\epsilon - \epsilon_\theta(\mathbf{y}_t, t, \mathbf{c})\|_2^2 \right], \quad (5)$$

where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\mathbf{y}_t = \sqrt{\bar{\alpha}_t} \mathbf{y}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$.

Algorithm 1 summarizes the reverse sampling procedure used to generate future velocity sequences during inference.

Train-only navigation field prior and local patching. We construct a train-only navigation field $\mathcal{F}$ that captures a coarse-grained mean flow of urban mobility in grid space. For each grid cell (i, j) , let $\mathcal{K}_{ij}$ be the set of training timesteps whose position falls into that cell. We estimate the mean step displacement and its direction as

$$\bar{\mathbf{v}}_{ij} = \frac{1}{|\mathcal{K}_{ij}|} \sum_{k \in \mathcal{K}_{ij}} \mathbf{v}_k, \quad \mathbf{u}_{ij} = \frac{\bar{\mathbf{v}}_{ij}}{\|\bar{\mathbf{v}}_{ij}\|_2 + \varepsilon}, \quad (6)$$

and record the mean speed $s_{ij} = \frac{1}{|\mathcal{K}_{ij}|} \sum_{k \in \mathcal{K}_{ij}} \|\mathbf{v}_k\|_2$ along with the visit count $n_{ij} = |\mathcal{K}_{ij}|$.

At inference time, we crop a local patch around the last observed position $\mathbf{p}_0$:

$$\Phi(\mathbf{p}_0) \in \mathbb{R}^{3 \times K \times K}, \quad \Phi = [\mathbf{u}_y, \mathbf{u}_x, s], \quad (7)$$

where $K = 32$ in our experiments. The patch is encoded by a lightweight CNN to produce an

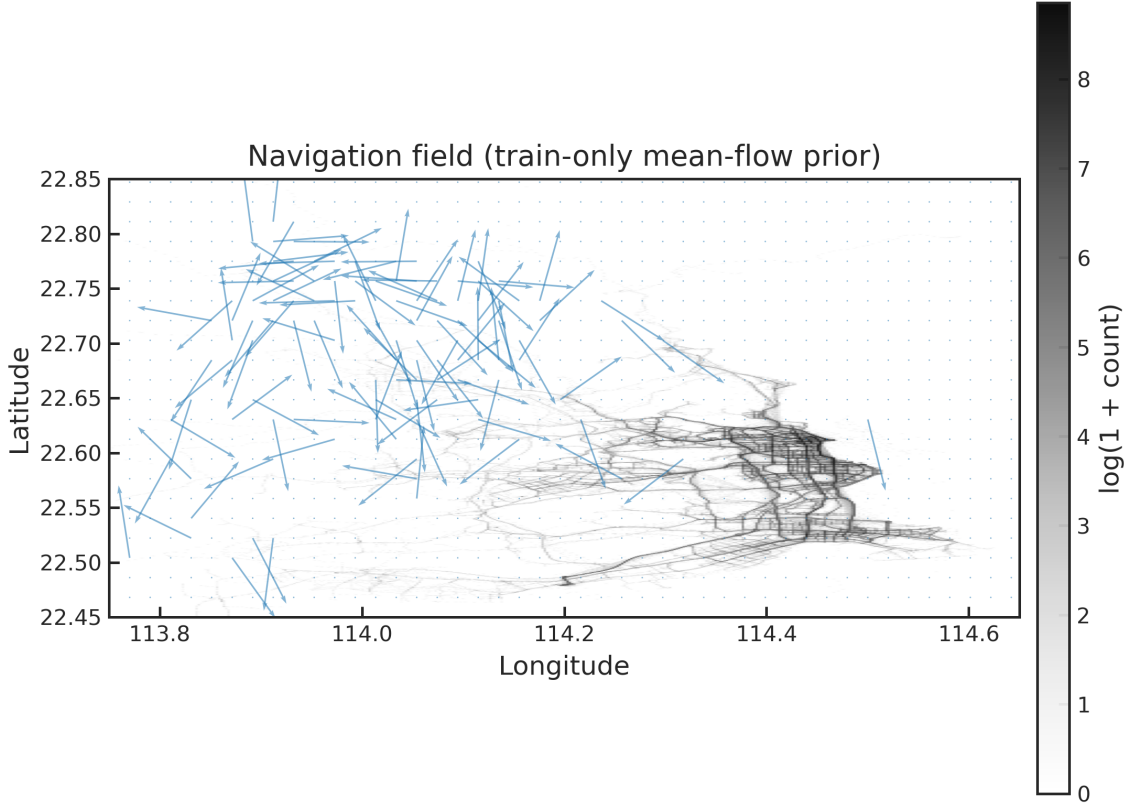


Figure 2: Train-only navigation field visualized in geographic space (Scheme B). Arrows indicate the local mean-flow direction, and the background intensity reflects log-scaled visit counts.

embedding $\mathbf{e}_{\text{nav}}$, which is concatenated with the global condition. Speed values are normalized by a train-only constant (`nav_max_speed`) to keep the patch scale stable.

Figure 2 visualizes the resulting train-only navigation field in geographic space; we use it as an interpretable mean-flow prior rather than a destination-specific shortest-path field.

3.3 Training and Evaluation

Training-time macro regularization (ablation). To mitigate the macroscopic “shrinkage” phenomenon (under-estimated net displacement), we explore an optional macro regularizer. Given $\mathbf{y}_t$ and the predicted noise ϵ_θ , we reconstruct an estimate of the clean sample

$$\hat{\mathbf{y}}_0 = \frac{\mathbf{y}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{y}_t, t, \mathbf{c})}{\sqrt{\bar{\alpha}_t} + 10^{-8}}. \quad (8)$$

We then compute the implied positions by summing displacements (Eq. 1) and apply an end-to-end displacement penalty:

$$\hat{\mathbf{p}}_k = \mathbf{p}_0 + \sum_{i=1}^k \hat{\mathbf{v}}_i, \quad \hat{\mathbf{v}} \in \hat{\mathbf{y}}_0, \quad (9)$$

$$\mathcal{L}_{\text{macro}} = \|(\hat{\mathbf{p}}_F - \mathbf{p}_0) - (\mathbf{p}_F - \mathbf{p}_0)\|_2^2. \quad (10)$$

To avoid applying geometric constraints at highly noisy diffusion timesteps, we gate this loss to early timesteps (higher signal-to-noise ratio): $\mathcal{L} = \mathcal{L}_{\text{diff}} + \lambda \mathbb{I}[t < \tau] \mathcal{L}_{\text{macro}}$.

Evaluation protocol. We evaluate on the test split using both micro-level and macro-level metrics. For generative models we draw $K = 20$ samples and report both mean and best-of- K (oracle) scores.

Micro metrics. We report ADE/FDE for each condition, and also compute discrete Fréchet distance and DTW as distribution-aware trajectory-shape measures. For ADE/FDE, we summarize **mean** over samples and **best-of- K** (minimum over samples) to quantify coverage.

Macro metrics. We report mean squared displacement (MSD) curves and radius of gyration (Rog) to characterize low-frequency motion. MSD is defined as

$$\text{MSD}(\tau) = \langle \|\mathbf{p}_{t+\tau} - \mathbf{p}_t\|_2^2 \rangle_{t, \text{conditions}}, \quad (11)$$

and Rog is computed per trajectory as $\text{Rog} = \sqrt{\frac{1}{F} \sum_{t=1}^F \|\mathbf{p}_t - \bar{\mathbf{p}}\|_2^2}$.

Geographic-space visualization (Scheme B). Models operate in grid space, but for urban interpretation we map grid coordinates back to latitude/longitude by a linear bbox projection using the recorded bbox and grid shape. This produces “map-like” overlays and density plots (without road-level map matching). Depending on the grid convention, the latitude axis may require a y-flip to match a north-up view; we follow the same convention as our visualization scripts and report this mapping as an approximate display transform.

4 Results

4.1 Phase B Results (dt-fixed=30s)

We report a quick but strict evaluation on 320 sampled conditions from the test split under dt-fixed=30s. Deterministic regression is evaluated with $K = 1$, while diffusion-based generators are evaluated with $K = 20$.

Model	K	ADE_{mean}	ADE_{best}	FDE_{mean}	FDE_{best}	MSD_{10}	Rog
Baseline (L2)	1	5.467	5.467	8.855	8.855	304.099	5.494
Diffusion	20	6.741	2.836	11.636	3.950	100.948	3.056
Physics	20	6.744	2.510	11.644	3.419	116.869	3.435

Table 1: Phase B quick results on the test split (320 conditions). Ground-truth macro references on the same conditions are $GT_Rog=5.247$ and $GT_MSD_{10}=349.740$.

Micro-level errors and coverage. The deterministic baseline achieves the best mean errors (Table 1), which is expected for L2 regression that approximates the conditional mean. In contrast, diffusion-based generators have higher mean errors but much stronger **best-of- K** capability: Physics improves the oracle upper bound over data-only diffusion (ADE_{best} 2.510 vs. 2.836; FDE_{best} 3.419 vs. 3.950), indicating better coverage when the navigation-field prior is provided.

Macro-level physical realism: shrinkage persists. Despite strong best-of- K , diffusion and physics-conditioned diffusion under-estimate macroscopic motion. For example, diffusion has $Rog=3.056$ compared to $GT_Rog=5.247$ (ratio ≈ 0.58), and MSD_{10} is 100.948 compared to $GT_MSD_{10}=349.740$ (ratio ≈ 0.29). Physics conditioning improves these macro statistics modestly ($Rog=3.435$, $MSD_{10}=116.869$) but does not eliminate the gap. Figure 3 visualizes this directly via micro-metric summaries and MSD curves over real-time lags. Additional qualitative and distributional summaries are provided in the Appendix (Figures 6–8).

Ablation: training-time macro supervision (EPE). Motivated by the shrinkage phenomenon, we tested a training-time macroscopic constraint based on end-to-end displacement error (Eq. 10). To avoid shortcut solutions at highly noisy diffusion timesteps, we apply hard gating (only for early timesteps) and use batch-level normalization for stability. Empirically, macro supervision is *sensitive*: if the weight is too small, improvements can plateau; if too large, the model may over-compensate and produce heavy-tailed motion that degrades *typical* samples ($K=1$), even if oracle best-of- K improves. This motivates a structural fix rather than further exhaustive λ sweeps.

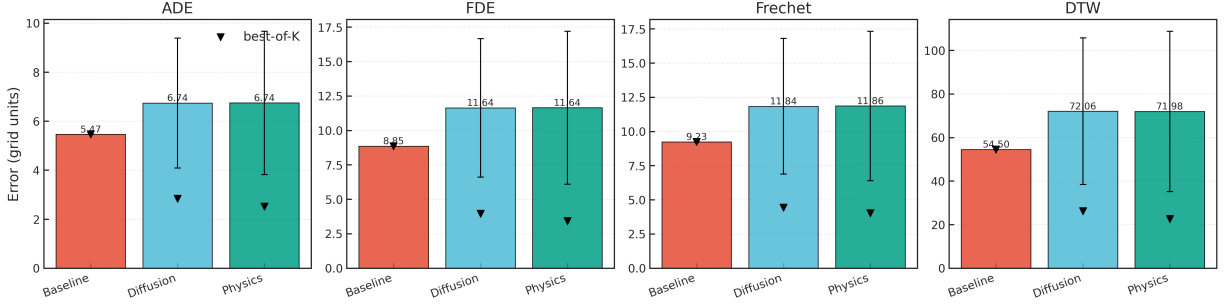
Structural fix: residual diffusion (v1.1). Given the limited gains from training-time macro loss, we adopt a residual formulation (Eq. 3) that uses a frozen deterministic baseline as a low-frequency prior and learns stochastic residuals with diffusion. Table 2 summarizes a fast, leakage-free check on the test split ($K = 10$) using a deterministic L2 baseline (trained under the same dt-fixed protocol) as a frozen low-frequency prior. Residual diffusion largely restores macroscopic scale (speed and Rog ratios close to 1) while maintaining strong oracle best-of- K capability. Residual *physics* diffusion further improves the oracle upper bound (best-of- K) but remains slightly conservative in

Model	K	ADE_{mean}	ADE_{best}	FDE_{mean}	FDE_{best}	Speed Ratio	Rog Ratio	MSD ₁₀ Ratio
Residual Diffusion	10	6.919	3.656	11.426	4.975	1.016	1.004	0.935
Residual Physics	10	5.948	2.709	9.885	3.569	0.951	0.945	0.864

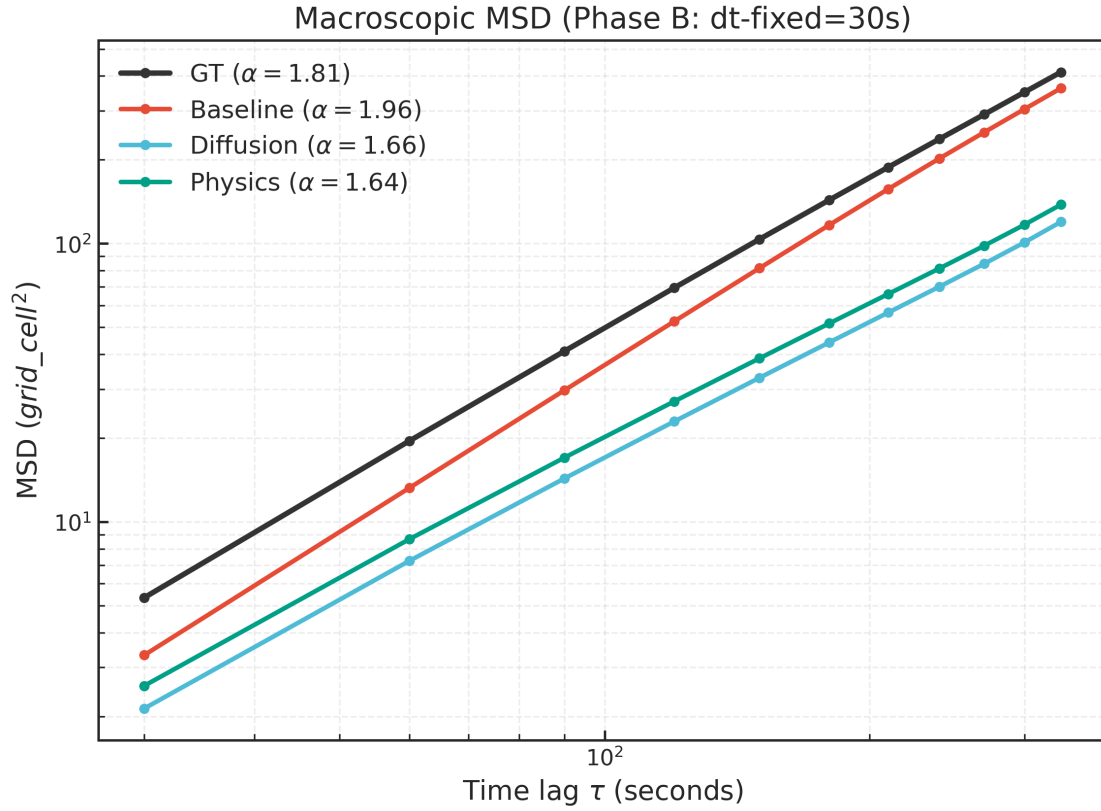
Table 2: Residual variants fast check on the test split ($K = 10$; subset evaluation). Ratios are computed against ground-truth statistics on the same evaluated subset under the same evaluation protocol.

MSD/Rog due to the mean-flow tether effect of the navigation-field prior. Unless otherwise stated, we keep the sampling schedule consistent with training (100 diffusion steps) and use smaller subsets (`max_batches`) only for quick validation; aggressive step-reduction can change calibration and is not used for reporting.

Interpretable geographic visualizations (Scheme B). To connect model behavior to urban spatial structure, we visualize a subset of saved evaluation samples ($N=80$ conditions) in geographic space using a linear bbox projection (without road-level map matching). Figure 4 shows trajectory overlays and Figure 5 shows spatial density maps. These “map-like” visualizations highlight (i) **spatial heterogeneity** (mobility concentrates along corridors and regions), and (ii) **probabilistic futures** (diffusion-based generators yield diverse continuations compared to a single mean path).

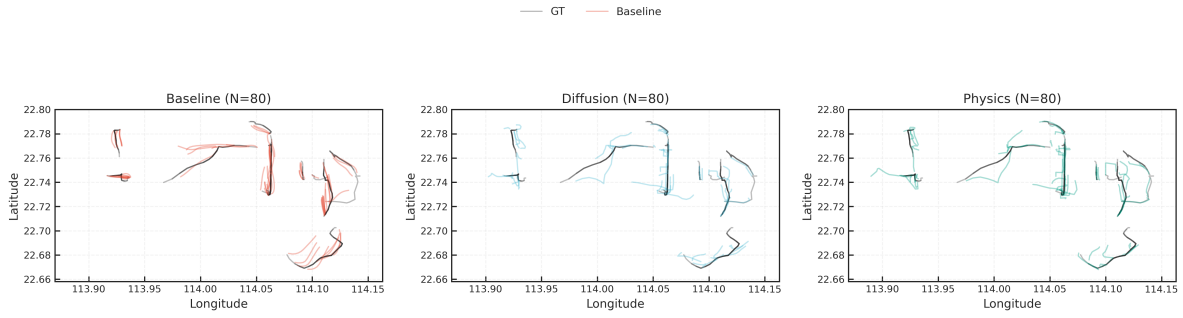


(a) Micro-level metrics (mean/std and best-of- K where applicable).

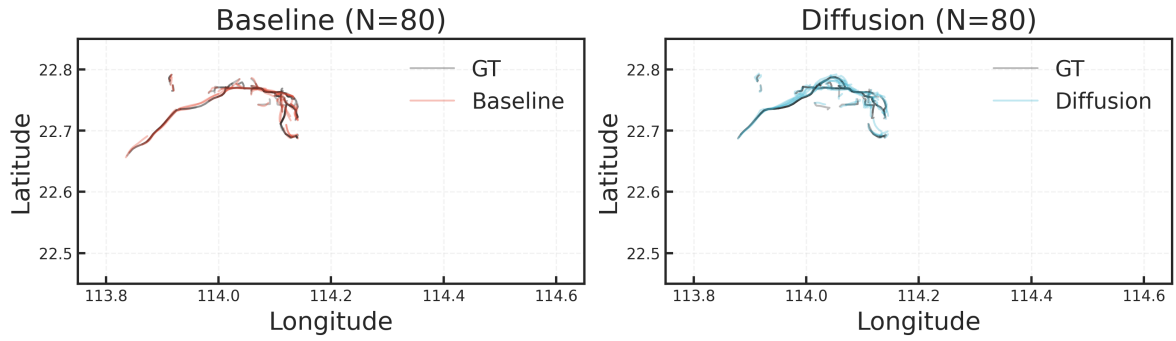


(b) MSD curves (Pred vs. GT) under dt-fixed=30s.

Figure 3: Phase B v1.0 quantitative evidence: diffusion-based generators improve best-of- K coverage but exhibit macroscopic shrinkage in MSD/Rog.

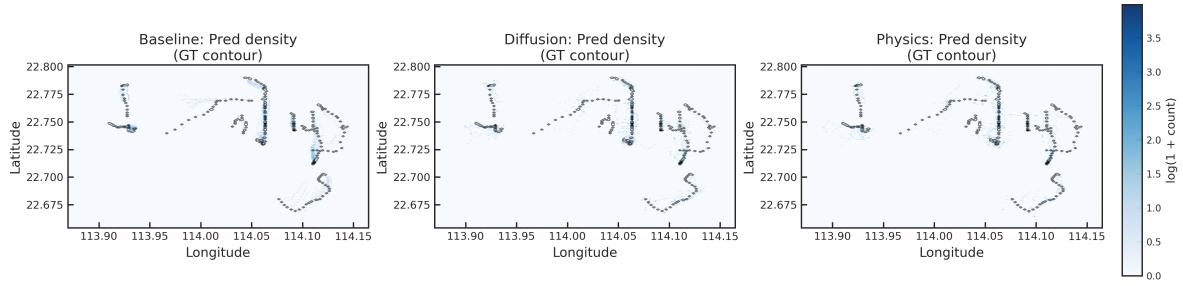


(a) Phase B v1.0: baseline vs diffusion vs physics (trajectory overlays).

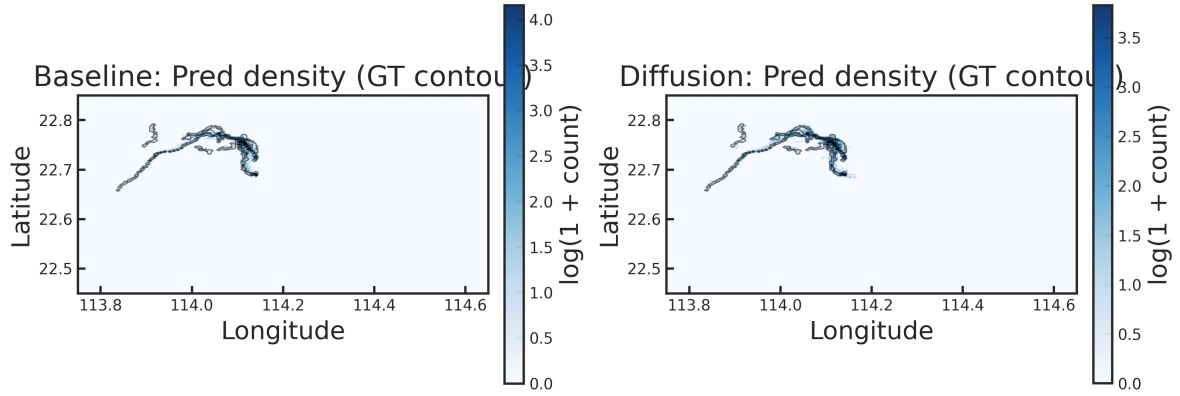


(b) Phase B v1.1: residual diffusion (frozen prior + stochastic residual).

Figure 4: Geographic-space visualizations (trajectory overlays) for urban interpretation.



(a) Phase B v1.0: predicted density with ground-truth contour reference.



(b) Phase B v1.1: density map under residual diffusion.

Figure 5: Geographic-space visualizations (density maps) for urban interpretation.

5 Discussion

Multi-scale tension: individual choice vs. collective flow. Our findings highlight a fundamental methodological tension in urban mobility modeling: the conflict between *microscopic fidelity* and *macroscopic consistency*. In traditional deterministic regression, the model acts as a “cautious planner”, collapsing the diverse potential of an urban grid into a single, mean-reverted path. While this minimizes point-wise error, it erases the spatial entropy—the branching of possibilities at intersections—that defines a living city. Denoising diffusion models successfully recover this entropy, as evidenced by their superior best-of- K coverage. However, they introduce a new risk: the “drift toward the mean” during iterative denoising, which manifests as macroscopic shrinkage.

Shrinkage as a geographic reflection of spatial entropy. The systematic under-estimation of displacement (low MSD/Rog) in Phase B v1.0 is more than a technical calibration error; it reflects a model’s struggle to maintain *directional persistence* across high-entropy urban junctions. At each denoising step, if the local navigation field or condition is ambiguous (e.g., a complex intersection), the generator tends to favor a conservative, low-velocity correction. Over multiple time steps, these cautious micro-adjustments accumulate into a macroscopic loss of momentum. Our analysis shows that post-hoc scaling is insufficient because it simply amplifies the model’s directional uncertainty.

A hierarchical multi-scale framework via residual diffusion. In Phase B v1.1, we demonstrate that a structural remedy—residual decomposition—outperforms naive loss regularization. By freezing a deterministic baseline as a low-frequency prior, we “anchor” the global scale of the trajectory. The diffusion process then focuses on stochastic residuals, i.e., plausible deviations from the mean flow. This creates a **hierarchical multi-scale framework**:

- **Macroscopic Anchor (Baseline):** Captures the fundamental OD-driven displacement and speed profile.
- **Mesosopic Guide (Nav Field):** Injects historical flow patterns and directional constraints.
- **Microscopic Variability (Diffusion Residual):** Learns the multi-modal distribution of human choice and local route variations.

This decomposition ensures that the generated futures are both diverse and physically grounded, bridging the gap between probabilistic coverage and macroscopic realism.

Implications for urban analytics and risk-aware planning. The value of this framework extends beyond academic metrics. For urban planners, the ability to generate *calibrated* multimodal futures

enables robust scenario testing. For instance, in emergency evacuation modeling or congestion management, knowing the “worst-case” dispersion or the probability of multiple route clusters can be more actionable than a single average prediction. Our geographic visualizations suggest that these models can reveal latent “flow corridors” and “spatial divergence zones” that are largely invisible to deterministic baselines.

Limitations and future work. Despite the success of the residual approach, our current implementation remains grid-based, which limits awareness of fine-grained road network topology. Future work should explore (i) road-graph-aware diffusion to better align with physical infrastructure, and (ii) dynamic coupling between the navigation field and real-time traffic signals. Furthermore, the sensitivity of macro-level statistics to the choice of the deterministic prior suggests that the “quality of the anchor” is a critical bottleneck. Exploring better-calibrated low-frequency priors—possibly physics-based kinematic models—remains a promising frontier for city-scale trajectory generation.

6 Conclusion

We have presented a physics-conditioned diffusion framework for origin–destination conditioned trajectory generation. By establishing a rigorous, leakage-free pipeline, we identified a fundamental tension in generative mobility modeling: the trade-off between local stochastic coverage and macroscopic physical consistency. Our diagnosis revealed that pure diffusion models struggle with “macroscopic shrinkage”—a failure to maintain net displacement in high-entropy urban environments.

To address this, we proposed a **Physics-Informed Residual Diffusion** framework that leverages a frozen deterministic prior to anchor the low-frequency scale. This approach improves macroscopic physical realism while preserving the rich, multimodal diversity of future paths. Our geographic visualizations confirm that this framework not only improves oracle coverage but also reproduces structured flow patterns of the city. Ultimately, this work provides a scalable, low-cost solution for recovering **simulation validity** in the next generation of urban Digital Twins.

7 Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this work, the authors used AI-assisted tools to support brainstorming, outlining, and technical writing refinement, as well as code/documentation review for experiment reproducibility. All generated suggestions were critically reviewed and edited by the authors. The authors take full responsibility for the final content, results interpretation, and conclusions.

8 Author Contribution Statement

Author: [Jinlin Wu]

My contributions. I contributed to: (i) defining the strict Phase B protocol (dt-fixed resampling, leakage-free data products and contracts), (ii) implementing and running training/evaluation pipelines (diffusion/physics models, metrics and ablations), (iii) analyzing results and diagnosing the shrinkage issue, and (iv) writing and editing this report, including figure preparation.

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A Appendix

A.1 Additional Quantitative Visualizations (Phase B)

To keep the main text focused, we place additional evaluation visualizations here. All plots are generated from the same strict Phase B pipeline (dt-fixed=30s) and use the saved evaluation samples where applicable.

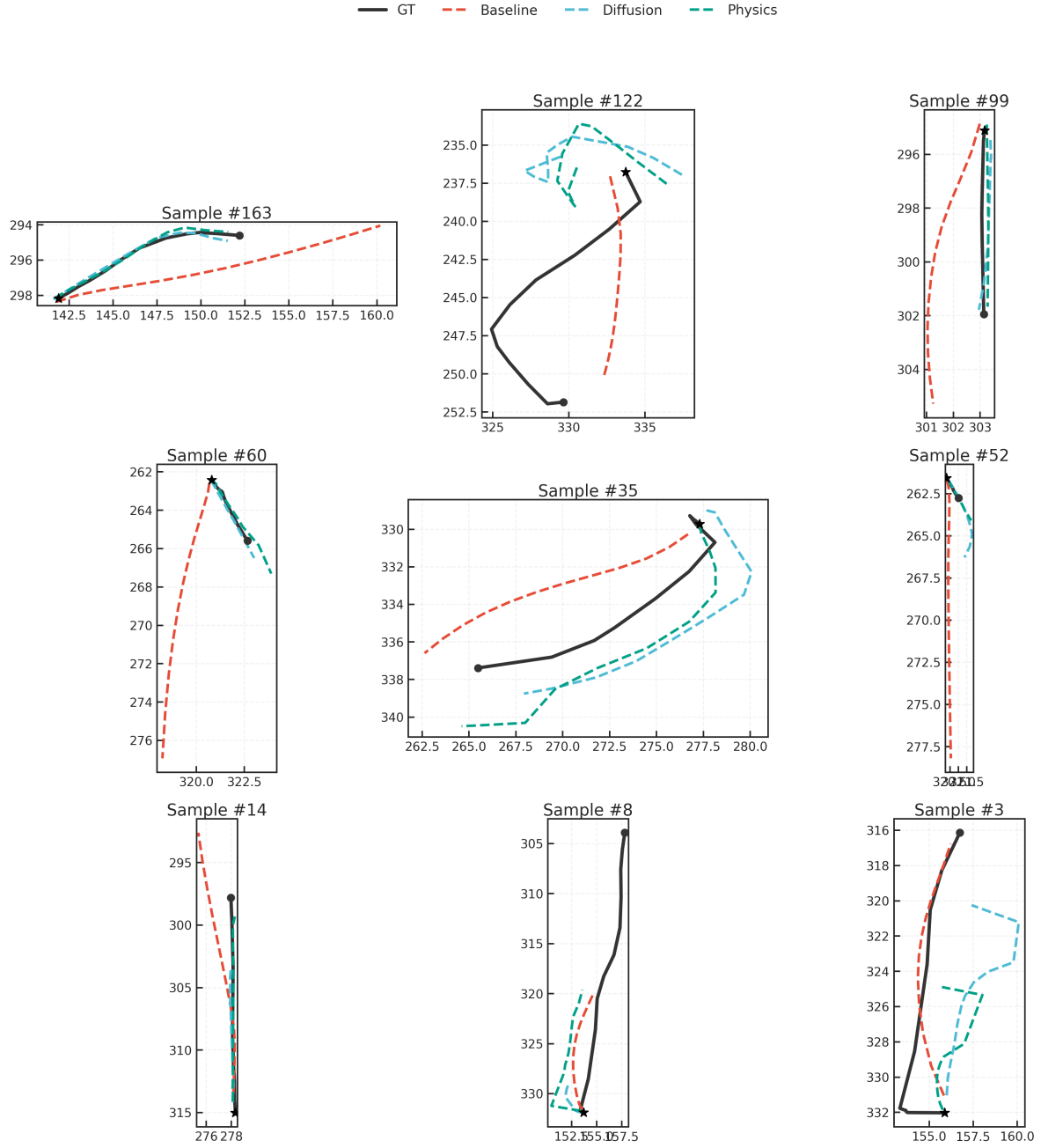


Figure 6: Qualitative trajectory overlays in grid space (multiple sampled conditions).

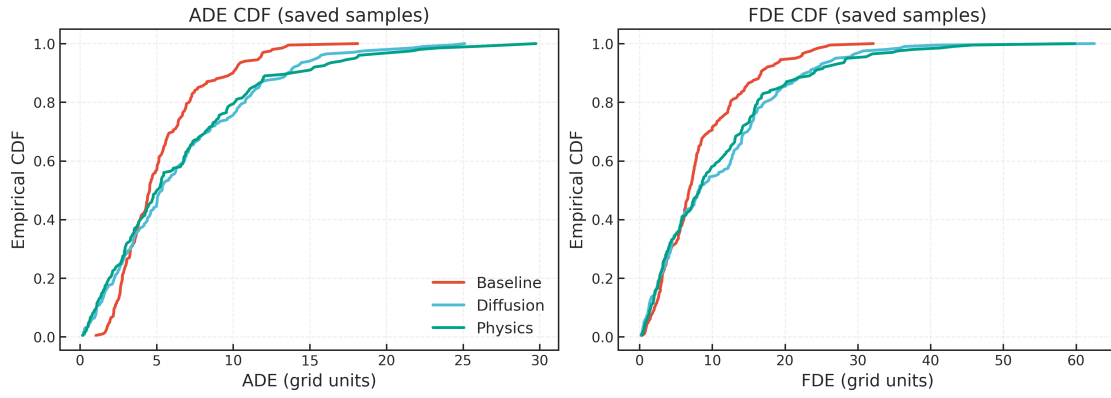


Figure 7: Empirical CDF of ADE/FDE computed on saved evaluation samples.

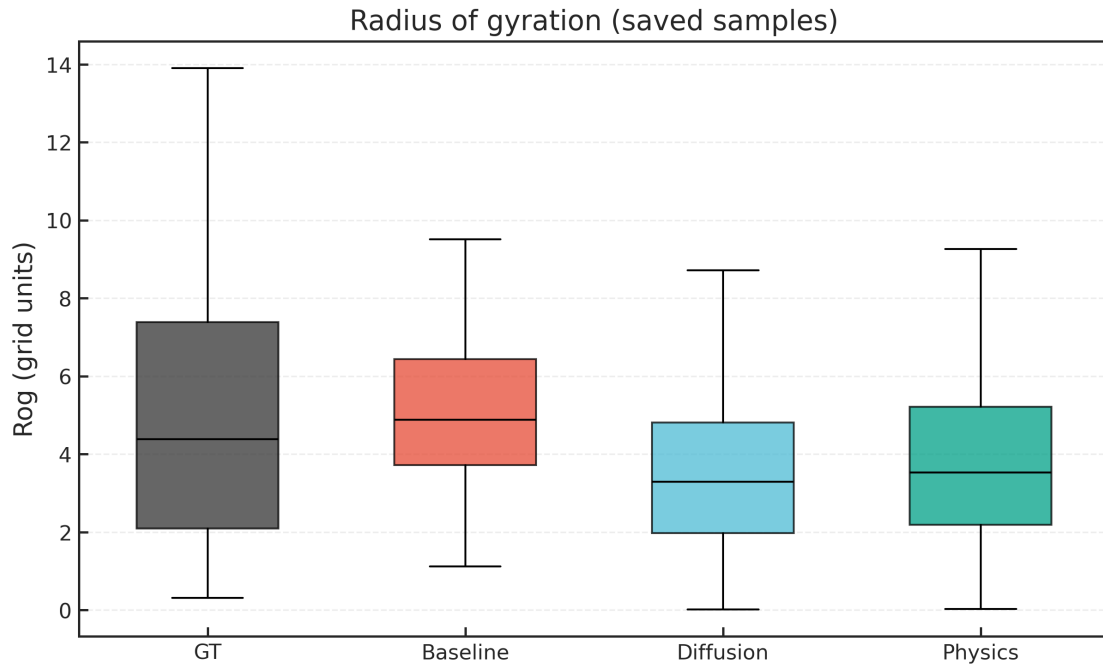


Figure 8: Distribution of radius of gyration (Rog) across trajectories (saved samples).